$A_3 = \{\langle M \rangle \mid M$ accepts 3 strings exactly $\}$.

$\boxed{\text{Theorem: } A_3 \text{ is undecidable.}}$

Suppose not. So, A_3 has a decider R (an oracle). We will show that HALT is decidable. Let $\langle M, w \rangle$ be an instance of HALT.

H = " On input $\langle M, w \rangle$:

Create the TM M'

M' = On input string x:

Run M on w.

If $x = 0$ or $x = 00$ or $x = 000$:

Accept

Else

Reject.

Use R with M' as input.

If R decides that M' accepts 3 strings, Accept. Otherwise, Reject. "

H is a decider for HALT, a contradiction. $\square$